

Solucion-tarea-13

Paso 0: Verificamos la conectividad de red

Comprobamos que los nodos se ven entre si con el comando ping -c

```
(kali㉿nodo1)-[~]  
$ ping -c 2 nodo2  
PING nodo2 (192.168.1.157) 56(84) bytes of data.  
64 bytes from nodo2 (192.168.1.157): icmp_seq=1 ttl=64 time=2.41 ms  
64 bytes from nodo2 (192.168.1.157): icmp_seq=2 ttl=64 time=4.47 ms  
  
— nodo2 ping statistics —  
2 packets transmitted, 2 received, 0% packet loss, time 1003ms  
rtt min/avg/max/mdev = 2.409/3.437/4.466/1.028 ms  
  
(kali㉿nodo1)-[~]  
$ ping -c 2 nodo3  
PING nodo3 (192.168.1.173) 56(84) bytes of data.  
64 bytes from nodo3 (192.168.1.173): icmp_seq=1 ttl=64 time=1.30 ms  
64 bytes from nodo3 (192.168.1.173): icmp_seq=2 ttl=64 time=2.99 ms  
  
— nodo3 ping statistics —  
2 packets transmitted, 2 received, 0% packet loss, time 1020ms  
rtt min/avg/max/mdev = 1.302/2.146/2.991/0.844 ms  
  
(kali㉿nodo1)-[~]  
$
```

Paso 1: Instalar Zookeeper

1. Descargamos zookeeper con el comando wget

```
(kali@nodo1)-[/tmp]
$ wget https://archive.apache.org/dist/zookeeper/zookeeper-3.8.3/apache-zookeeper-3.8.3-bin.tar.gz -O /tmp
--2026-04-29 10:55:11-- https://archive.apache.org/dist/zookeeper/zookeeper-3.8.3/apache-zookeeper-3.8.3-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 85.180.204.109, 2a01:4f9:1a:a004::2
Connecting to archive.apache.org (archive.apache.org)[85.180.204.109:443]... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14853274 (14M) [application/x-gzip]
Saving to: '/tmp/apache-zookeeper-3.8.3-bin.tar.gz'

apache-zookeeper-3.8.3-bin.tar.gz 100%[=====>] 14.17M 299KB/s in 51s
2026-04-29 10:56:08 (284 KB/s) - '/tmp/apache-zookeeper-3.8.3-bin.tar.gz' saved [14853274/14853274]
```

2. Extraemos lo descargado con el comando tar -xzf

```
(kali@nodo1)-[/tmp]
$ tar -xzf /tmp/apache-zookeeper-3.8.3-bin.tar.gz -C /tmp
```

3. Movemos a /usr/local con el comando sudo mv

```
(kali@nodo1)-[/tmp]
$ sudo mv /tmp/apache-zookeeper-3.8.3-bin /usr/local/zookeeper
[sudo] password for kali:
```

4. Verificamos con el comando ls a la carpeta donde lo hemos movido

```
(kali@nodo1)-[/tmp]
$ ls /usr/local/zookeeper/bin
README.txt      zkCli.cmd       zkEnv.cmd       zkServer.cmd     zkServer.sh     zkSnapshotComparer.sh  zkSnapshotToolkit.sh  zkTxnLogToolkit.sh
zkCleanup.sh    zkCli.sh        zkEnv.sh        zkServer-Initialize.sh  zkSnapshotComparer.cmd  zkSnapshotToolkit.cmd  zkTxnLogToolkit.cmd
```

Paso 2: Configurar Zookeeper, para ello necesitamos crear el archivo de configuración

1. Copiamos el archivo de configuración que viene de ejemplo con el comando `sudo cp`

```
(kali㉿ nodo1)-[/tmp]
$ sudo cp /usr/local/zookeeper/conf/zoo_sample.cfg /usr/local/zookeeper/conf/zoo.cfg
```

2. Editamos el archivo con el comando `sudo nano` lo abrimos. Al final del archivo añadimos unas líneas para definir los tres nodos del clúster.

```
GNU nano 8.6 /usr/local/zookeeper/conf/zoo.cfg
# The number of milliseconds of each tick
tickTime=2000
# The number of ticks that the initial
# synchronization phase can take
initLimit=10
# The number of ticks that can pass between
# sending a request and getting an acknowledgement
syncLimit=5
# the directory where the snapshot is stored.
# do not use /tmp for storage, /tmp here is just
# example sakes.
dataDir=/tmp/zookeeper
# the port at which the clients will connect
clientPort=2181
# the maximum number of client connections.
# increase this if you need to handle more clients
#maxClientCnxns=60
#
# Be sure to read the maintenance section of the
# administrator guide before turning on autopurge.
#
# https://zookeeper.apache.org/doc/current/zookeeperAdmin.html#sc_maintenance
#
# The number of snapshots to retain in dataDir
#autopurge.snapRetainCount=3
# Purge task interval in hours
# Set to "0" to disable auto purge feature
#autopurge.purgeInterval=1

## Metrics Providers
#
# https://prometheus.io Metrics Exporter
#metricsProvider.className=org.apache.zookeeper.metrics.prometheus.PrometheusMetricsProvider
#metricsProvider.httpHost=0.0.0.0
#metricsProvider.httpPort=7000
#metricsProvider.exportJvmInfo=true

server.1=nodo1:2888:3888
server.2=nodo2:2888:3888
server.3=nodo3:2888:3888
```

3. Creamos el directorio de datos y asignamos la identidad de nodo1, posteriormente comprobamos que todo se haya ejecutado correctamente con el comando `cat`

```
(kali㉿ nodo1)-[/tmp]
$ mkdir -p /tmp/zookeeper && echo "1" > /tmp/zookeeper/myid

(kali㉿ nodo1)-[/tmp]
$ cat /tmp/zookeeper/myid
1
```

4. Arrancamos zookeeper en cada nodo con el comando start y comprobamos si todo ha ido bien con el comando status

```
(kali@nodo1)-[~]
$ /usr/local/zookeeper/bin/zkServer.sh start
/usr/bin/java
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED

(kali@nodo1)-[~]
$ /usr/local/zookeeper/bin/zkServer.sh status
/usr/bin/java
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: follower

(kali@nodo1)-[~]
$
```

```
(kali@nodo2)-[~]
$ /usr/local/zookeeper/bin/zkServer.sh start
/usr/bin/java
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED

(kali@nodo2)-[~]
$ /usr/local/zookeeper/bin/zkServer.sh status
/usr/bin/java
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: follower

(kali@nodo2)-[~]
$
```

```
(kali@nodo3)-[~]
$ /usr/local/zookeeper/bin/zkServer.sh start
/usr/bin/java
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED

(kali@nodo3)-[~]
$ /usr/local/zookeeper/bin/zkServer.sh status
/usr/bin/java
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: leader

(kali@nodo3)-[~]
$
```

PASO 3

1. Descargamos Hbase con el comando wget

```
(kali@nodo1:~)$ wget https://archive.apache.org/dist/hbase/2.4.17/hbase-2.4.17-bin.tar.gz -P /tmp
--2026-04-30 07:18:18-- https://archive.apache.org/dist/hbase/2.4.17/hbase-2.4.17-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.180.204.189, 2a01:4f01:a804::2
Connecting to archive.apache.org (archive.apache.org)|65.180.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20485851 (272M) [application/gzip]
Saving to: '/tmp/hbase-2.4.17-bin.tar.gz'

hbase-2.4.17-bin.tar.gz          100%[>] 271.66M  294kB/s  in 21m 3s
2026-04-30 07:19:22 (220 KB/s) - '/tmp/hbase-2.4.17-bin.tar.gz' saved [20485851/20485851]
```

2. Extraemos con el comando tar -xzf

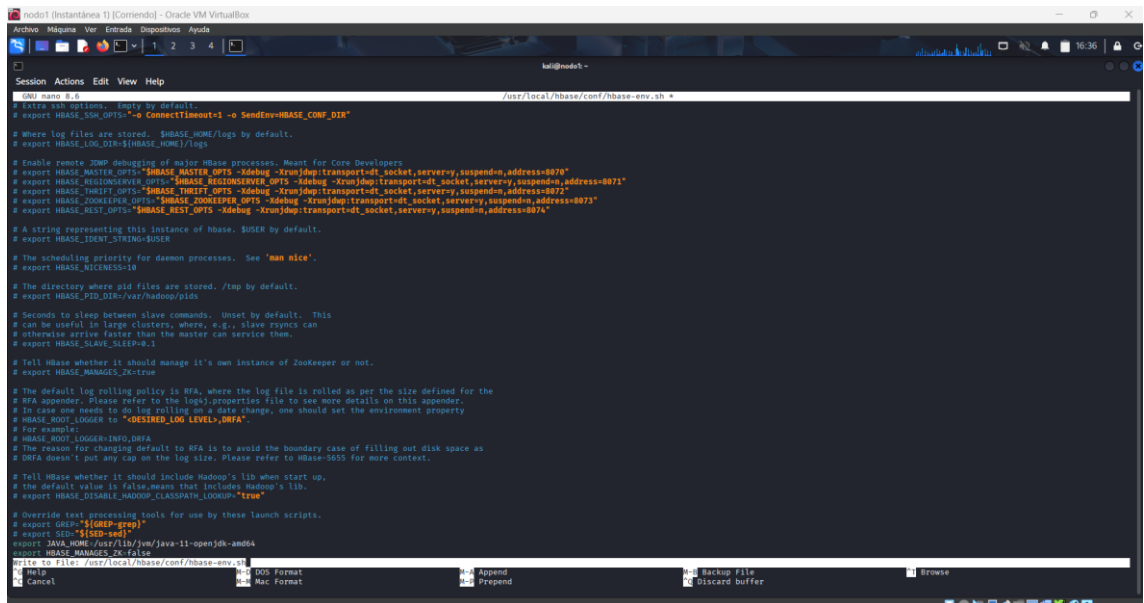
```
(kali@nodo1:~)$ tar -xzf /tmp/hbase-2.4.17-bin.tar.gz -C /tmp
```

3. Movemos a /usr/local con el comando mv y verificamos que todo se ha hecho correctamente

```
(kali@nodo1:~)$ sudo mv /tmp/hbase-2.4.17 /usr/local/hbase
[sudo] password for kali:

(kali@nodo1:~)$ ls /usr/local/hbase
bin  CHANGES.md  conf  docs  hbase-webapps  LEGAL  lib  LICENSE.txt  NOTICE.txt  RELEASNOTES.md
```

4. Editamos el archivo hbase-env.sh con añadiéndole al final unas líneas



```
GNU nano 2.9.6 /usr/local/hbase/conf/hbase-env.sh
# Set HBase configuration. Empty by default.
# export HBASE_SSH_OPTS="-o ConnectTimeout=1 -o SendEnv=HBASE_CONF_DIR"

# Where log files are stored. $HBASE_HOME/logs by default.
# export HBASE_LOG_DIR=${HBASE_HOME}/logs

# Enable remote JMX debugging of major HBase processes. Meant for Core Developers
# export HBASE_MASTER_OPTS="-Xmx1G -Xms128M -Xloggc:/tmp/hbase-master.log -XX:+UseG1GC -XX:MaxDirectMemorySize=1024M"
# export HBASE_REGIONSERVER_OPTS="-Xmx1G -Xms128M -Xloggc:/tmp/hbase-regionserver.log -XX:+UseG1GC -XX:MaxDirectMemorySize=1024M"
# export HBASE_THRIFT_OPTS="-Xmx1G -Xms128M -Xloggc:/tmp/hbase-thrift.log -XX:+UseG1GC -XX:MaxDirectMemorySize=1024M"
# export HBASE_ZOOKEEPER_OPTS="-Xmx1G -Xms128M -Xloggc:/tmp/hbase-zookeeper.log -XX:+UseG1GC -XX:MaxDirectMemorySize=1024M"
# export HBASE_REST_OPTS="-Xmx1G -Xms128M -Xloggc:/tmp/hbase-rest.log -XX:+UseG1GC -XX:MaxDirectMemorySize=1024M"

# A string representing this instance of hbase. $USER by default.
# export HBASE_IDENT_STRING=$USER

# The scheduling priority for daemon processes. See 'man nice'.
# export HBASE_NICE=10

# The directory where pid files are stored. /tmp by default.
# export HBASE_PID_DIR=/var/hadoop/pids

# Seconds to sleep between slave commands. Unset by default. This
# can be useful in large clusters, where, e.g., slave syncs can
# otherwise arrive faster than the master can service them.
# export HBASE_SLAVE_SLEEP=0.1

# Tell HBase whether it should manage it's own instance of Zookeeper or not.
# export HBASE_MANAGES_ZK=true

# The default log rolling policy is RFA, where the log file is rolled as per the size defined for the
# RFA appender. Please refer to the logs/properties file to see more details on this appender.
# In case one needs to do log rolling on a date change, one should set the environment property
# HBASE_ROOT_LOGGER to "com.red_helix.log.level,DRFA".
# For example:
# HBASE_ROOT_LOGGER=INFO,DRFA
# The reason for changing default to RFA is to avoid the boundary case of filling out disk space as
# DRFA doesn't put any cap on the log size. Please refer to Hbase-555 for more context.

# Tell HBase whether it should include Hadoop's lib when start up.
# the default value is false, means that includes Hadoop's lib.
# export HBASE_DISABLE_HADOOP_CLASSPATH_LOOKUP=true

# Override test processing tools for use by these launch scripts.
# export GDB="gdb"
# export SED="sed"
# export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
# export HBASE_MANAGES_ZK=false
```

5. Editamos el archivo hbase-site.xml

```
Session Actions Edit View Help
GNU nano 2.6 /usr/local/hbase/conf/hbase-site.xml

# regarding copyright ownership. The ASF licenses this file
# to you under the Apache License, Version 2.0 (the
# "License"); you may not use this file except in compliance
# with the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
<configuration>
<!--
The following properties are set for running Hbase as a single process on a
developer workstation. With this configuration, Hbase is running in
"stand-alone" mode and without a distributed file system. In this mode, and
without further configuration, Hbase and Zookeeper data are stored on the
local filesystem, in a path under the value configured for 'hbase.tmp.dir'.
This value is overridden from its default value of '/tmp' because many
systems clean /tmp on a regular basis. Instead, it points to a path within
this Hbase installation directory.

Running against the 'LocalFileSystem', as opposed to a distributed
filesystem, runs the risk of data integrity issues and data loss. Normally
Hbase will refuse to run in such an environment. Setting
'hbase.unsafe.stream.capability.enforce' to 'false' overrides this behavior,
permitting operation. This configuration is for the developer workstation
only and __should not be used in production!__

See also https://hbase.apache.org/book.html#standalone_dist
-->
<property>
<name>hbase.rootdir</name>
<value>hdfs://nod01:9000/hbase/</value>
</property>
<property>
<name>hbase.cluster.distributed</name>
<value>true</value>
</property>
<property>
<name>hbase.zookeeper.quorum</name>
<value>nod01,nod02,nod03</value>
</property>
</configuration>

Help Write Out Where Is Cut Execute Location Undo Set Mark To Bracket Previous Back Prev Word Home
Exit Read File Replace Paste Justify Go To Line Redo Copy Where Was Next Forward Next Word End
```

6. Editamos el archivo regionserver

```
nod01 (Instantánea 1) [Corriendo] - Oracle VM VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda
1 2 3 4
kali@nod01:~$
Session Actions Edit View Help
GNU nano 2.6 /usr/local/hbase/conf/regionserver
nod02
nod03
```

7. Copiamos Hbase a nodo2 y nodo3 con el comando scp -r

```
[kali@nod01]~$
[~$ scp -r /usr/local/hbase kali@nod02:/tmp/

jetty-6.1.26.jar 100% 527KB 12.0MB/s 00:00
junit-4.13.2.jar 100% 376KB 30.2MB/s 00:00
mockito-core-2.28.2.jar 100% 577KB 35.5MB/s 00:00
hamcrest-core-1.3.jar 100% 48KB 0.9MB/s 00:00
jakarta.validation-api-2.0.2.jar 100% 90KB 19.5MB/s 00:00
hadoop-yarn-server-timelineservice-2.10.0.jar 100% 198KB 23.0MB/s 00:00
curator-client-2.0.0.jar 100% 291KB 31.4MB/s 00:00
jakarta.inject-2.6.1.jar 100% 18KB 5.3MB/s 00:00
hadoop-mapreduce-client-common-2.10.0.jar 100% 771KB 27.4MB/s 00:00
jettison-1.3.4.jar 100% 88KB 12.4MB/s 00:00
msql-jdbc-6.2.1.jre7.jar 100% 774KB 25.5MB/s 00:00
disruptor-3.4.2.jar 100% 81KB 14.4MB/s 00:00
hadoop-minicluster-2.10.0.jar 100% 39KB 7.2MB/s 00:00
commons-net-3.1.jar 100% 267KB 14.9MB/s 00:00
byte-buddy-agent-1.9.10.jar 100% 42KB 9.5MB/s 00:00
hbase-shaded-miscellaneous-4.1.4.jar 100% 3982KB 24.0MB/s 00:00
hbase-endpoint-2.4.17.jar 100% 460KB 19.4MB/s 00:00
hadoop-annotations-2.10.0.jar 100% 46KB 7.0MB/s 00:00
hbase-server-2.4.17.jar 100% 528KB 23.2MB/s 00:00
guice-servlet-3.0.jar 100% 63KB 10.5MB/s 00:00
hadoop-yarn-server-resourcemanager-2.10.0.jar 100% 191KB 17.0MB/s 00:00
zookeeper-jute-3.5.7.jar 100% 245KB 16.4MB/s 00:00
commons-collections-3.2.2.jar 100% 575KB 13.0MB/s 00:00
jndi-api-2.2.1.jar 100% 18KB 6.5MB/s 00:00
jetty-servlet-6.1.26.jar 100% 1671KB 19.5MB/s 00:00
jcodings-1.0.55.jar 100% 226KB 32.1MB/s 00:00
hbase-shell-2.4.17.jar 100% 1175KB 36.4MB/s 00:00
hadoop-mapreduce-client-core-2.10.0.jar 100% 158KB 22.3MB/s 00:00
snappy-java-1.0.5.jar 100% 1222KB 31.5MB/s 00:00
jclap-1.0.5.jar 100% 4722 1.0MB/s 00:00
hadoop-yarn-server-applicationhistoryservice-2.10.0.jar 100% 529KB 14.9MB/s 00:00
hadoop-mapreduce-client-hs-2.10.0.jar 100% 195KB 18.0MB/s 00:00
woodstox-core-6.0.3.jar 100% 581KB 23.3MB/s 00:00
netty-handler-4.1.45.Final.jar 100% 439KB 35.9MB/s 00:00
jackson-core-2.14.1.jar 100% 449KB 21.0MB/s 00:00
hbase-metrics-2.4.17.jar 100% 28KB 7.4MB/s 00:00
curator-recipes-2.6.1.jar 100% 289KB 18.3MB/s 00:00
hadoop-yarn-server-web-proxy-2.10.0.jar 100% 66KB 10.9MB/s 00:00
hbase-shaded-client-bye-hadoop-2.4.17.jar 100% 498B 20.9MB/s 00:00
hbase-shaded-mapreduce-2.10.0.jar 100% 318B 15.9MB/s 00:00
asn-1.1.jar 100% 42KB 4.3MB/s 00:00
fit-2.58.jar 100% 279KB 10.2MB/s 00:00
hbase-logging-2.4.17.jar 100% 13KB 1.0MB/s 00:00
listdrift@1a.1.jar 100% 258KB 0.8MB/s 00:00
hbase-rest-2.4.17.jar 100% 434KB 7.3MB/s 00:00
hadoop-yarn-client-2.10.0.jar 100% 278KB 6.3MB/s 00:00
```

```
[kali@nod01]~$
[~$ scp -r /usr/local/hbase kali@nod03:/tmp/

jetty-6.1.26.jar 100% 527KB 12.0MB/s 00:00
junit-4.13.2.jar 100% 376KB 30.2MB/s 00:00
mockito-core-2.28.2.jar 100% 577KB 35.5MB/s 00:00
hamcrest-core-1.3.jar 100% 48KB 0.9MB/s 00:00
jakarta.validation-api-2.0.2.jar 100% 90KB 19.5MB/s 00:00
hadoop-yarn-server-timelineservice-2.10.0.jar 100% 198KB 23.0MB/s 00:00
curator-client-2.0.0.jar 100% 291KB 31.4MB/s 00:00
jakarta.inject-2.6.1.jar 100% 18KB 5.3MB/s 00:00
hadoop-mapreduce-client-common-2.10.0.jar 100% 771KB 27.4MB/s 00:00
jettison-1.3.4.jar 100% 88KB 12.4MB/s 00:00
msql-jdbc-6.2.1.jre7.jar 100% 774KB 25.5MB/s 00:00
disruptor-3.4.2.jar 100% 81KB 19.2MB/s 00:00
hadoop-minicluster-2.10.0.jar 100% 39KB 7.2MB/s 00:00
commons-net-3.1.jar 100% 267KB 15.5MB/s 00:00
byte-buddy-agent-1.9.10.jar 100% 42KB 9.5MB/s 00:00
hbase-shaded-miscellaneous-4.1.4.jar 100% 3982KB 24.0MB/s 00:00
hbase-endpoint-2.4.17.jar 100% 460KB 19.4MB/s 00:00
hadoop-annotations-2.10.0.jar 100% 46KB 7.0MB/s 00:00
hbase-server-2.4.17.jar 100% 528KB 23.2MB/s 00:00
guice-servlet-3.0.jar 100% 63KB 10.5MB/s 00:00
hadoop-yarn-server-resourcemanager-2.10.0.jar 100% 191KB 17.0MB/s 00:00
zookeeper-jute-3.5.7.jar 100% 245KB 16.4MB/s 00:00
commons-collections-3.2.2.jar 100% 575KB 13.0MB/s 00:00
jndi-api-2.2.1.jar 100% 18KB 6.5MB/s 00:00
jetty-servlet-6.1.26.jar 100% 1671KB 19.5MB/s 00:00
jcodings-1.0.55.jar 100% 226KB 32.1MB/s 00:00
hbase-shell-2.4.17.jar 100% 1175KB 36.4MB/s 00:00
hadoop-mapreduce-client-core-2.10.0.jar 100% 158KB 22.3MB/s 00:00
snappy-java-1.0.5.jar 100% 1222KB 31.5MB/s 00:00
jclap-1.0.5.jar 100% 4722 733.6KB/s 00:00
hadoop-yarn-server-applicationhistoryservice-2.10.0.jar 100% 529KB 14.9MB/s 00:00
hadoop-mapreduce-client-hs-2.10.0.jar 100% 195KB 18.0MB/s 00:00
woodstox-core-6.0.3.jar 100% 581KB 23.3MB/s 00:00
netty-handler-4.1.45.Final.jar 100% 439KB 35.9MB/s 00:00
0K 0 0.0MB/s --- ETA*
100% 195KB 15.2MB/s 00:00
100% 581KB 7.3MB/s 00:00
100% 439KB 23.3MB/s 00:00
```

8. Movemos los archivos de hbase a la carpeta correcta

```
(kali@nodo2)-[~]  
$ sudo mv /tmp/hbase /usr/local/hbase  
[sudo] password for kali:
```

```
(kali@nodo3)-[~]  
$ sudo mv /tmp/hbase /usr/local/hbase  
[sudo] password for kali:
```

9. Arrancamos Hbase desde el nodo1

```
(kali@nodo1)-[~]  
$ /usr/local/hbase/bin/start-hbase.sh  
running master, logging to /usr/local/hbase/bin/../logs/hbase-kali-master-nodo1.out  
nodo2: running regionserver, logging to /usr/local/hbase/bin/../logs/hbase-kali-regionserver-nodo2.out  
nodo3: running regionserver, logging to /usr/local/hbase/bin/../logs/hbase-kali-regionserver-nodo3.out  
  
(kali@nodo1)-[~]  
$
```

10. Explotar datos con Hbase Shell

```
(kali@nodo1)-[~]  
$ /usr/local/hbase/bin/hbase shell  
2026-04-30 17:08:52,429 WARN [main] util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable  
WARNING: An illegal reflective access operation has occurred  
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (file:/usr/local/hbase/lib/hadoop-auth-2.10.0.jar) to method sun.security.krb5.Config.getInstance()  
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authentication.util.KerberosUtil  
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations  
WARNING: All illegal access operations will be denied in a future release  
HBase Shell  
Use "help" to get list of supported commands.  
Use "exit" to quit this interactive shell.  
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell  
Version 2.4.17, r7fd996f39b4284da9a71da3ce67c48d259ffa79a, Fri Mar 31 18:10:45 UTC 2023  
Took 0.0039 seconds  
hbase:001:0>
```

11. Creamos una tabla

```
hbase:002:0> create 'alumnos', 'info'  
Created table alumnos  
Took 1.2913 seconds  
⇒ Hbase::Table - alumnos  
hbase:003:0>
```

12. Insetamos un registro con el comando put

```
hbase:002:0> create 'alumnos', 'info'  
Created table alumnos  
Took 1.2913 seconds  
⇒ Hbase::Table - alumnos  
hbase:003:0> put 'alumnos', 'id_01', 'info:nombre', 'Manu'  
Took 1.3382 seconds  
hbase:004:0> scan 'alumnos'  
ROW COLUMN+CELL  
id_01 column=info:nombre, timestamp=2026-05-01T06:45:18.807, value=Manu  
1 row(s)  
Took 0.0962 seconds  
hbase:005:0>
```

PASO 4: Spark sobre Yarn

1. Descargamos Spark con el comando wget

```
(kali@nodo1: ~)
$ wget https://archive.apache.org/dist/spark/spark-3.4.2/spark-3.4.2-bin-hadoop3.tgz -P /tmp
--2026-05-01 06:47:33-- https://archive.apache.org/dist/spark/spark-3.4.2/spark-3.4.2-bin-hadoop3.tgz
Resolving archive.apache.org (archive.apache.org)... (65.100.204.100, 2001:4911a:a0b0::2)
Connecting to archive.apache.org (archive.apache.org)|65.100.204.100|:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 388664780 (370M) [application/gzip]
Saving to: '/tmp/spark-3.4.2-bin-hadoop3.tgz'

spark-3.4.2-bin-hadoop3.tgz 100%[=====] 370.60M 6.65MB/s in 65s
2026-05-01 06:48:39 (5.67 MB/s) - '/tmp/spark-3.4.2-bin-hadoop3.tgz' saved [388664780/388664780]
```

2. Extraemos con el comando tar -xzf

```
(kali@nodo1)~$ tar -xzf /tmp/spark-3.4.2-bin-hadoop3.tgz -C /tmp
```

3. Movemos a /usr/local con el comando sudo mv

```
(kali@nodo1)~$ sudo mv /tmp/spark-3.4.2-bin-hadoop3 /usr/local/spark
[sudo] password for kali:
```

4. Añadimos las variables de entorno

```
GNU nano 8.6 /home/kali/.bashrc
alias ll='ls -l'
alias la='ls -A'
alias l='ls -CF'

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this; if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -o posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi

# Java
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$JAVA_HOME/bin:$PATH

# Hadoop
export HADOOP_HOME=/home/kali/Desktop/hadoop
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
export HADOOP_CLASSPATH=$JAVA_HOME/lib/tools.jar

# HIVE
export HIVE_HOME=/home/kali/Desktop/hive4
export PATH=$PATH:$HIVE_HOME/bin

export SQOOP_HOME=/opt/hadoop/sqoop
export PATH=$PATH:$SQOOP_HOME/bin

export SPARK_HOME=/usr/local/spark
export PATH=$PATH:$SPARK_HOME/bin
```

5. Iniciamos spark

```
Using Python version 3.13.7 (main, Aug 20 2025 22:17:40)
Spark context Web UI available at http://nodo1:4040
Spark context available as 'sc' (master = yarn, app id = application_1777629735094_0001).
SparkSession available as 'spark'.
>>> datos = sc.parallelize(range(1, 100))
>>> print(datos.map(lambda x: x * 2).sum())
9900
>>>
```